

EGS3012S: Atmospheric Dynamics & Thermodynamics Glossary 2026

Stefaan Conradie

August 16, 2026

Contents

1	Introduction	1
2	Glossary	2
2.1	Climate system	2
2.2	Energy balance	3
2.3	Thermodynamics	3
2.4	Dynamics	5
3	Reference list of mathematical symbols	7
4	Description of mathematical symbols	8
5	Acronyms	11
6	Units	12
7	Formula sheet (to be provided in tests and exams)	13

1 Introduction

Here I provide a brief description of some key concepts ([section 2](#)) that we will cover atmospheric dynamics & thermodynamics section of the course (what I'm teaching). In [section 3](#), an explanation of mathematical

symbols used in equations in this course is provided; more details for mathematical quantities are provided in [section 4](#). Commonly used acronyms are included in [section 5](#), followed by the course formula sheet ([section 7](#)). In addition to the descriptions provided here, also see the following two links to open resources, that provide descriptions of most of these concepts and many others relevant to the course.

See also:

Glossary of the American Meteorological Society

IPCC glossary The glossary of the Intergovernmental Panel on Climate Change, as used for its latest 6th cycle assessment report (AR6). The IPCC is a panel of climate and earth system scientists assembled by the United Nations to summarise the state of knowledge on observed and projected climate changes, its impacts and possible approaches to mitigate and adapt to it.

2 Glossary

2.1 Climate system

weather The instantaneous, local condition of the atmosphere and short-term variation therein. Weather timescales typically range from seconds to days. Weather describes the here and now, particularly in relation to temperature, precipitation (rainfall, snow, hail, dew, etc.), humidity, wind (speed & direction) and sunshine/cloudiness.

climate A longer-term statistical, or aggregate description of weather tendencies. Climate is intended to tell us what sort of weather we can expect. It covers likely weather and a range possible weather conditions, including extremes, for particular seasons, times of day or night and areas. Climate can cover timescales from seasons to millions of years. Climate descriptions or statistics also often cover larger areas, from cities to ocean basins or the whole planet. Often climate is described using 30-year monthly average data, but this is an arbitrary convention.

atmosphere The thin layer of gasses that surrounds the solid Earth (or another planet). The gasses are prevented from leaking out into space by the planet's gravity. The atmosphere is composed of various layers, the lowest and (for our purposes) most important of which is the **troposphere**.

climate system The complex collection of interacting and interrelated components whose behaviour strongly influence the state of the atmosphere on longer timescales and thereby determine its response to externally-driven changes. The components of the climate system are generally regarded as:

- the **atmosphere**, which varies mostly on shorter timescales, up to a few weeks.
- the **oceans**, lakes and rivers, collectively the **hydrosphere**, which varies primarily on timescales from months to centuries (the deep ocean)
- the **cryosphere**, all the **ice** on Earth's surface, mostly sea ice, land ice, glaciers and snow pack. Ice, of course, is water in its solid state, so the cryosphere is a part of the hydrosphere, but is often treated separately, because of its different albedo and other properties.
- the **geosphere**, or the lithosphere, the land surface and soils

- the **biosphere**, all life on the Earth's surface, considered primarily from the perspective of chemical exchanges, albedo and moisture **fluxes** (transfers between different components of the climate system).

2.2 Energy balance

insolation or **incoming solar radiation**. High-energy, short-wave electromagnetic radiation received from the Sun. Insolation heats the Earth's surface, which absorbs the insolation, before radiating out **long-wave terrestrial radiation** (lower energy), which heats the atmosphere from below.

electromagnetic radiation Energy moving through space as a combination of perpendicular waves in electric and magnetic fields. Visible light is a form of electromagnetic magnetic radiation. The longer the wavelength of radiation the lower the frequency and associated energy of the radiation. This is the form in which energy is received and emitted by the Earth system.

greenhouse gas (GHG) Also referred to as "radiatively active gases".

radiation Energy propagated through space by electric and magnetic fields, radiated out by objects, including atmospheric particles, based on their temperature and emissivity. Includes visible light

conduction Energy transfer through a medium or between substances in contact with one another due to random motion and consequent collisions between the molecular particles that constitute the media or substances.

convection Vertical transport of heat and moisture (and other atmospheric properties) in a fluid (chiefly the atmosphere) due to buoyancy differences or actions forcing vertical displacement of fluid volumes

sensible heat Transfer of energy between bodies in contact with each other that alters their temperatures without changing the phase of water

latent heat Heat released or required for a change of phase of water at constant temperature

scattering The action of atmospheric particles causing radiation to change direction

absorption Energy taken on by a substance or object due to transfer of energy to it

reflection For radiation, a process whereby an object redirects the motion of energy away from the surface of the body, as though the radiation bounces off the the object.

2.3 Thermodynamics

air column Theoretical convenience, like an imaginary cylinder we place in the atmosphere to isolate air with similar properties from the surroundings

air parcel Theoretical convenience; a volume or bubble of air that we can treat as having uniform properties throughout, such as temperature, humidity, density, etc.

buoyancy The tendency of a fluid element (e.g. water or air) to rise if it is less dense than surrounding fluid.

isobar Lines connecting points in space with equal pressure

isotherm Lines connecting points in space with equal temperature

isohume Lines connecting points in space with equal humidity (e.g. mixing ratio or vapour pressure)

isoline Lines connecting points in space with equal variable values

contour Lines connecting points in space of equal height, also geopotential height

lifting condensation level (LCL) The height or pressure level to which an air parcel with a given temperature and humidity would need to be lifted adiabatically to reach saturation due to cooling. Generally the height at which clouds would form in an air column where moist air is lifted from the surface.

level of free convection (LFC) The height or pressure level to which an air parcel would need to be lifted to subsequently continue rising by itself, due to it being warmer and thus more buoyant than the surrounding environmental profile

equilibrium level (EL) The height or pressure level above the LFC at which the temperature of a lifted air parcel will be equal to the environmental temperature again and thus would no longer continue to rise. This often represents the top of convective clouds.

convective available potential energy (CAPE) The energy available to a thunderstorm due to the greater buoyancy of a lifted air parcel relative to its environment. Usually measured between the LFC and EL. An air parcel needs to be lifted to the LFC first before CAPE is available

convective inhibition (CIN) Energy required to lift an air parcel to the LFC

saturation When air holds as much water vapour as it can at a given pressure and temperature, the saturation vapour pressure is reached (equivalently, the saturated mixing ratio and specific humidity, 100% relative humidity and zero dew point depression). At this point, the processes of evaporation and condensation are in equilibrium and the relative amounts of liquid and gaseous water in the air does not change.

subsaturated or unsaturated Air parcels where the vapour pressure is lower than the saturated vapour pressure as determined by the Clausius Clapeyron relation. Evaporation dominates condensation, so liquid water drops in such an environment shrink in size.

supersaturated Air parcels where the vapour pressure is greater than the saturated vapour pressure as determined by the Clausius Clapeyron relation. In such parcels condensation occurs more rapidly than condensation. Hence, if $T \gtrsim 0^\circ\text{C}$, liquid water droplets tend to grow in size at the expense of water vapour; below freezing, ice crystals will tend to grow at the expense of vapour.

dew point depression Difference between the parcel temperature and dew point temperature: $T - T_d$

atmospheric stability An atmospheric profile wherein a parcel in the environment perturbed from its initial position will tend to return towards the initial position when released

atmospheric instability An atmospheric profile wherein a parcel lifted in the environment continues to rise further

conditional instability An atmospheric profile wherein a saturated air parcel is unstable and an unsaturated air parcel is stable

adiabatic An adiabatic process does not involve a parcel exchanging heat or energy with its surroundings

inversion layer A slice or layer of the atmosphere in which temperature, contrary to expectation in the troposphere, increases with increasing height and decreasing pressure.

potential temperature The temperature an air parcel would have if it were adiabatically warmed by taking it to where $P = 1000\text{hPa}$ (near sea-level)

equivalent potential temperature The temperature an air parcel would have if it were adiabatically warmed by taking it to where $P = 1000\text{hPa}$, after all its water vapour had been condensed out.

virtual temperature The temperature that a dry air parcel would have if its pressure and density were equal to those of a moist air parcel of interest. Can be used instead of T in the dry air equation of state ($P = \rho RT$) to make the equation approximately valid for moist air.

2.4 Dynamics

pressure gradient force (PGF) The force experienced by a parcel of air as a result of differences in atmospheric pressure at the same height in the atmosphere. Air tends to be pushed away from areas of high pressure towards areas of low pressure. Alternatively, you can think of it as the force experienced due to the air parcels along the same pressure surface being at different heights and tending to slide downwards on the pressure surface.

Coriolis (pseudo-)force An apparent deflection of air due to the differential rate of rotation of the Earth at different latitudes and its spherical geometry. The strength of the effect is $2\Omega \sin \phi$ and it acts to the right in the northern hemisphere and left in the southern hemisphere.

centripetal force A force acting inwards towards the centre of a circle thus maintaining circular motion

centrifugal force An apparent force acting where isobars are curved, due to the inertial tendency of an air parcel or fluid element to continue in a straight line at constant speed

friction A force tending to retard the motion of an object, body or fluid element

cyclone A large-scale closed low-pressure system, where pressure is lower than the surrounding areas

anti-cyclone A large-scale closed high-pressure system, where pressure is greater than the surrounding areas

trough An elongated region of lower pressure, without closed isobars

ridge An elongated region of higher pressure, without closed isobars

geostrophic flow The theoretical wind that would result from a perfect balance between only the Coriolis and PGF

gradient wind The theoretical wind that would result from a perfect balance between the Coriolis, Centrifugal and PGF

ageostrophic wind or flow The difference between the theoretical geostrophic wind at a given location and the observed or actual wind at that location

convergence Air movement in towards an area, leading to mass accumulation

divergence Air movement out away from an area, leading to mass depletion

non-divergence Where neither convergence nor divergence occurs

wind shear A change in wind in space or time, e.g. vertical wind shear is a change of wind with height above the surface

thermal wind Vertical wind shear explained by the horizontal thickness gradient between air columns of differential (virtual) temperature, as quantified by the hypsometric equation.

backing An anti-clockwise change in wind direction

veering A clockwise change in wind direction

cold advection When winds transport air from a colder to a warmer location

warm advection When winds transport air from a warmer to a colder location

barotropic atmosphere Where isotherms are parallel to isobars

thermally direct circulation Where warm air rises and cold air sinks

thermally indirect circulation Where warm sinks and cold air rises, due to dynamic processes

jet streak Regions of maximum wind speeds within a jetstream

cut-off low A cold-cored closed low in the mid-latitude mid-troposphere, cut off and isolated from the major westerly wind belt

group velocity The speed at which a wave pattern, rather than the air within the wave, moves

air masses Large relatively uniform volumes/masses of air, in terms of properties such as temperature and moisture

weather front A sharp, dynamic boundary between different air masses with different densities

absolute vorticity The sum of planetary and relative vorticity

teleconnection A process whereby the weather and climate of a remote region is influenced by a mode of variability occurring in a particular region of the world, e.g. the connection between African rainfall variability and the El Niño-Southern Oscillation (ENSO)

3 Reference list of mathematical symbols

symbol	standard unit	variable name	further notes
S	$\text{W} \cdot \text{m}^{-2}$	solar constant	Average radiative flux at Earth's orbital radius
T	K	air temperature	Also referred to as “dry bulb temperature”
T_p	K	parcel temperature	
Θ	K	potential temperature	not to be confused with θ (longitude angle)
m	kg	mass	
$\perp, \parallel$		perpendicular/parallel to	
ρ	$\text{kg} \cdot \text{m}^{-3}$	density	mass per unit volume ($= \frac{m}{V}$)
P	$\text{Pa} = \text{N} \cdot \text{m}^{-2}$	pressure	normal force per unit area ($= \frac{F_{\perp}}{A}$)
F	$\text{N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$	force	$= ma$
D	m	depth	height of a column of air
A	$\text{m}^2; (\text{m})$	area (or wave amplitude)	
V	m^3	volume	
σ	$\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$	Stefan-Boltzmann constant	
λ	m	wavelength	distance from peak-to-peak along a wave
ν	s^{-1}	wave frequency	wave crests passing per unit time
c	$\text{m} \cdot \text{s}^{-1}$	wave speed	$c = \lambda\nu$
e, e_s	Pa	(saturated) vapour pressure	partial pressure due to water vapour
q	unitless ($\text{kg} \cdot \text{kg}^{-1}$)	specific humidity	vapour mass divided by air mass
X	unitless ($\text{kg} \cdot \text{kg}^{-1}$)	mixing ratio	vapour mass divided by dry air mass
α	unitless	albedo	proportion of received energy reflected by a surface
ϵ	unitless	emissivity	efficiency of a body at emitting radiation
x, y, z	m	position	co-ordinates of a point in space
n	m	normal distance	$\perp$ to isobars = $\parallel$ to pressure gradient
r	m	radius (of curvature)	distance from the centre of a circle,
θ	unitless/ $^{\circ}\text{E}$	longitude	angular distance from Prime Meridian
ϕ	unitless/ $^{\circ}\text{N}$	latitude	angular distance from Equator
a	$\text{m} \cdot \text{s}^{-2}$	acceleration	rate of change of speed/velocity $\vec{a} = \frac{d\vec{v}}{dt}$

symbol	standard unit	variable name	further notes
u, v, w	$\text{m} \cdot \text{s}^{-1}$	x, y, z wind components	vector components describing air movement
$ \vec{V} $	$\text{m} \cdot \text{s}^{-1}$	wind speed	magnitude of horizontal wind vector
ψ	unitless (radian)	wind direction; zenith angle	for wind: bearing angle where wind comes from
$\vec{V}_g$	$\text{m} \cdot \text{s}^{-1}$	geostrophic wind	theoretical wind balancing Coriolis and PGF
$\vec{V}_{\text{gr}}$	$\text{m} \cdot \text{s}^{-1}$	gradient wind	adds Centrifugal force to the geostrophic balance
f, ζ	s^{-1}	planetary, relative vorticity	$\hat{\mathbf{k}} \cdot \nabla \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$
Div	s^{-1}	horizontal divergence	$\nabla_H \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$

4 Description of mathematical symbols

Position

x Position or displacement along the zonal (West–East) direction. The positive direction is East.

y Position or displacement along the meridional (South–North) direction. The positive direction is North.

z Position or displacement along the vertical (Down–Up) direction. The positive direction is up.

Mass

P Atmospheric pressure, the force per unit surface area exerted by the mass of all air above the surface. Pressure is exerted simultaneously in all directions. Often used to denote a pressure level in the vertical direction.

P_0 Reference atmospheric pressure; unless otherwise indicated $= 1000\text{hPa} = 10^5\text{Pa}$

Z Geopotential height, the height in the atmosphere at which column air pressure would reach a defined level, such as $P = 500\text{hPa}$, if gravitational acceleration g were independent of height above the Earth’s surface. Z at 500hPa is denoted as Z_{500} .

Wind

u Wind vector component along the zonal (West–East) direction. Components give the direction air is flowing in (towards) not from, like direction.

v Wind vector component along the meridional (South–North) direction.

w Velocity component along the vertical (Down–Up) direction. The positive direction is up.

$|\vec{V}|$ Wind speed. Using Pythagoras, in 2-D: $|\vec{V}| = \sqrt{u^2 + v^2}$, in 3-D $|\vec{V}| = \sqrt{u^2 + v^2 + w^2}$.

$u_g, v_g, |\vec{V}_g|$ Geostrophic wind component along the zonal (West–East) and meridional (South–North) direction and the corresponding magnitude (horizontal wind speed).

$u_{gr}, v_{gr}, |\vec{V}_{gr}|$ Gradient wind component along the zonal (West–East) and meridional (South–North) direction and the corresponding magnitude.

$w_{\max}$ Maximum theoretically attainable updraft speed in a thunderstorm environment with a given CAPE value

Angles

ψ Wind direction bearing, indicating where the wind is coming *from*. The bearing angle measured clockwise from North (i.e. the positive y -axis), not like in Maths, where you measure anti-clockwise from East.

θ Longitude, usually in degrees (range is $[-180^\circ, 180^\circ]$). Degrees East are positive, West negative.

ϕ Latitude. Usually in degrees (range is $[-90^\circ, 90^\circ]$). Degrees North are positive, South negative.

Vorticity and Divergence

f Coriolis parameter. Defined as $2\Omega \sin(\phi)$. Also called **planetary vorticity**.

Ω Rate of angular rotation of the Earth = $\frac{2\pi}{86400} \approx 7.29 \times 10^{-5} \text{ s}^{-1}$.

ζ Relative vorticity, a measure of the rotation of an air parcel relative to the Earth's surface; more precisely, the vertical component of the curl of the velocity field

Div Horizontal divergence, also just called divergence, a measure of outward motion of air and thus mass depletion. Negative divergence, indicating inward motion and mass accumulation, is called **convergence**

Temperature

T_e Environmental temperature, i.e. temperature of air surrounding an air parcel.

Θ Potential temperature; the temperature an air parcel would have if adiabatically moved to $P_0 = 1000 \text{ hPa}$. Note that θ is used for angles, including **longitude**. Conserved under adiabatic processes when saturation is not reached.

Θ_{eq} Equivalent potential temperature, the temperature an air parcel would have if all moisture had been condensed out (and the released latent heat used to warm the air parcel) and then adiabatically warmed through. Conserved under

Γ Temperature lapse rate = $-\frac{\partial T}{\partial z}$, the rate at which air temperature decreases with height

Γ_e Environmental lapse rate, the observed lapse rate in an air column

Γ_d Dry adiabatic lapse rate, the theoretical process lapse rate for an unsaturated air parcel moving vertically without heat exchange with the environment

Γ_s Moist/Wet/Saturated adiabatic lapse rate, as Γ_d , for a saturated air parcel

Moisture

T_d Dew-point temperature, the temperature to which an air parcel will need to be cooled to reach saturation while keeping specific humidity (equivalently vapour pressure) fixed.

T_w Wet-bulb temperature, the temperature to which an air parcel can be cooled by evaporating moisture into it

T_v Virtual temperature of a moist air parcel M , the temperature a dry air parcel would have to have the same density as the M at the same pressure

q Specific humidity, the ratio of the mass of water vapour to the total mass of air

q_s Saturation specific humidity, the specific humidity value at which saturation is reached for a given temperature and pressure.

e Vapour pressure, the contribution to total air pressure contributed by water vapour, conventionally reported in hPa. Not to be confused with the constant $e = 2.7182818...$, the base of the natural logarithm.

e_0 The saturated vapour pressure at the freezing point of water (0°C), measured as 611.3hPa

RH Relative humidity($\%$) = $\frac{100e}{e_s}$

X Mixing ratio, the ratio of the mass of water vapour to the total mass of dry air in an air parcel

η The ratio of the molar mass of water vapour to that of dry air = 0.622

C_p The heat capacity of a substance, i.e. the amount of energy required to increase the temperature of 1 unit mass of the substance by 1K. SI units: $\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$

$C_{p,d}$ The heat capacity of dry air

Energy

E_{OLR} Rate of outgoing long-wave energy transfer per unit area from a body, with SI units of $\text{W} \cdot \text{m}^{-2}$

S Solar constant $\approx 1367 \text{W} \cdot \text{m}^{-2}$

CAPE Convective available potential energy; SI units: $\text{J} \cdot \text{kg}^{-1}$

CIN Convective inhibition; SI units: $\text{J} \cdot \text{kg}^{-1}$

Φ^* Net radiative flux between Earth's surface and atmosphere

Φ_H Surface sensible heat flux

Φ_E Surface latent heat flux (positive for evaporation, negative for condensation)

Φ_G Molecular heat conduction within the surface (ground/ocean) layer

Constants

σ The Stephan-Boltzmann constant, which relates the temperature of a surface with the energy it emits.

$$\sigma = 5.67 \times 10^{-8} \text{ K } (\cdot \text{W} \cdot \text{m}^{-2})^{-1}$$

R Specific gas constant for dry air. $R = 287 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$

R_v Specific gas constant for water vapour. $R_v = 461 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$

C Wien's Law constant. $C = 2897 \text{ } \mu\text{m K} = 2.897 \times 10^{-3} \text{ m K}$

g Acceleration due to gravity near the Earth's surface. $g \approx 9.8 \text{ m} \cdot \text{s}^{-2}$

R_E Radius of the Earth $\approx 6.4 \times 10^6 \text{ m}$

L_v Latent heat of vapourisation; energy required per unit mass of liquid water to convert the liquid to a gas (water vapour); $L_v = 2.5 \times 10^6 \text{ J} \cdot \text{kg}^{-1}$

Parameters

α Albedo of a surface (proportion of received energy which is reflected).

k Wavenumber; the number of waves of a given wavelength that fit into a latitude circle. If the wavelength in degrees longitude of a planetary zonal wave is λ , the corresponding wavenumber is $k = \frac{360^\circ}{\lambda}$.

β Bowen Ration $\beta = \frac{\Phi_H}{\Phi_E}$ (ratio of sensible to latent heat fluxes). Also sometimes used for the rate of change of the Coriolis parameter with meridional distance $\frac{df}{dy}$

5 Acronyms

SLP Sea-Level Pressure

OLR Outgoing long-wave radiation

GHG Greenhouse gas

NH, SH Northern & Southern Hemispheres

LCL Lifting condensation level

LFC Level of free convection

EL Equilibrium level

CAPE Convective available potential energy

CIN Convective inhibition

MSE Moist Static Energy

DSE Dry Static Energy

CCN Cloud condensation nuclei

IN Ice nuclei

DALR Dry adiabatic lapse rate, Γ_d

SALR Saturated adiabatic lapse rate, Γ_d

ELR environmental lapse rate, Γ_d

PGF Pressure gradient force

ENSO El Niño-Southern Oscillation

SAM/ AAO Southern Annular Mode/ Antarctic Oscillation

IOD Indian Ocean Dipole

RH Relative humidity

COL Cut-Off low

AR Atmospheric river

MLC Mid-Latitude cyclone

TC Tropical cyclone

LP, HP Low and high-pressure cells

SAHP, SIHP South Atlantic and Indian Ocean High Pressure cells

ITCZ Inter-Tropical Convergence Zone

SW, SE, NW, NE Inter-Cardinal directions: south-west, south-east, north-west and north-east

ABL Atmospheric Boundary Layer

SST Sea surface temperature

DJF, MAM, JJA, SON The meteorological austral summer (December-February), autumn, winter and spring, uniquely identified by the sequence of first letters of the months that make up the season; can be used for any season of 2 months or more, e.g. AM = April-May and ONDJFM is the extended summer semester: October–March.

6 Units

Units and unit-conversion are crucial in any natural science, including climatology and meteorology. For equations that are derived from Physics laws and principles (e.g. $F = ma$), unless you can cancel other units out, it is safest to first convert all quantities you use to SI units (for our purposes mostly: metre (m), second (s), kilogram (kg), Kelvin (K), Pascal (Pa) and combinations of these). Quantities input into logarithms (such as $\ln$) or exponents (like x in $e^x = \exp(x)$) must always be dimensionless (i.e. “unitless”).

7 Formula sheet (to be provided in tests and exams)

$$\begin{aligned}
P &= \rho R T_v & \Phi^* &= \Phi_G - \Phi_H - \Phi_E & \frac{\partial P}{\partial z} &= -\rho g & \Delta z &\approx \frac{R}{g} \bar{T}_v \ln \left(\frac{P_1}{P_2} \right) \\
E_{\text{OLR}} &= \epsilon \sigma T^4 & \sigma &= 5.67 \times 10^{-8} \text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4} \\
\lambda_{\text{max}} &= \frac{C}{T} & C &= 2897 \mu\text{m} \cdot \text{K} = 2.897 \times 10^{-3} \text{m} \cdot \text{K} \\
g &= 9.8 \text{m} \cdot \text{s}^{-2} & R &= 287 \text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1} & S &= 1367 \text{W} \cdot \text{m}^{-2} & \rho_0(T=288\text{K}, P=1013\text{hPa}) &= 1.22 \text{kg} \cdot \text{m}^{-3} \\
P &= \frac{F_{\perp}}{A} \approx \frac{mg}{A} & \rho &= \frac{m}{V} & \Gamma &= -\frac{\partial T}{\partial z} & P_2 &= P_1 \left(\frac{T_2}{T_1} \right)^b & b &= \frac{g}{\Gamma R} \\
e_s &= e_0 \exp \left(\frac{L_v}{R_v} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right); & T_0 &= 0^\circ\text{C} = 273.15\text{K} & e_0 &= e_s(T=0^\circ\text{C}) = 611.3\text{Pa} \\
L_v &= 2.50 \times 10^6 \text{J} \cdot \text{kg}^{-1} & R_v &= 461 \text{J} \cdot \text{K}^{-1} \cdot \text{kg}^{-1}, & C_{p,d} &= 1004 \text{J} \cdot \text{kg}^{-1} \text{K}^{-1} \\
\Delta E_d &= g \Delta Z + C_{p,d} \Delta T & \Delta E_s &\approx g \Delta Z + C_{p,d} \Delta T + L_v \Delta q \\
q &= \frac{(\eta)(e)}{P - (e)(1 - \eta)} = \frac{X}{1 + X} & e &= \frac{(X)(P)}{\eta + X} & X &= \frac{(\eta)(e)}{P - e} & \eta &= \frac{M_{\text{H}_2\text{O}}}{M_{\text{air}}} = 0.622 \\
T_v &= T \left(\frac{1 + \frac{X}{\eta}}{1 + X} \right) \approx T (1 + 0.61X) & \text{RH} &= 100 \frac{e}{e_s} \approx 100 \frac{q}{q_s} \approx \frac{100X}{X_s} \text{ in } \% \\
\Theta &= T + \Gamma_d Z = T \left(\frac{P_0}{P} \right)^{\frac{R}{C_{p,d}}} & \frac{R}{C_{p,d}} &= 0.286; & P_0 &= 1000 \text{hPa} \\
\text{CAPE} &= g \int_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z & w_{\text{max}} &= \sqrt{2(\text{CAPE})} \\
\text{CIN} &= g \int_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z \\
\Theta_{\text{eq}} &\approx \Theta + \frac{(L_v)(\Theta)(X)}{(C_{p,d})(T)} & Z_{\text{LCL}} &\approx 125(T - T_d) \text{ in metres}
\end{aligned}$$

here: $\exp(x) \equiv e^x$; Unless otherwise stated, assume $\Gamma_d = 0.010 \text{K} \cdot \text{m}^{-1}$ and $\Gamma_s = 0.006 \text{K} \cdot \text{m}^{-1}$

For dynamics:

$$\begin{aligned}
\psi &= \begin{cases} 90^\circ - \arctan\left(\frac{v}{u}\right) & u < 0 \\ 270^\circ - \arctan\left(\frac{v}{u}\right) & u > 0 \end{cases} & |\vec{V}| &= \sqrt{u^2 + v^2} & u &= -|\vec{V}| \sin(\psi) & v &= -|\vec{V}| \cos(\psi) \\
|\vec{V}_g| &= \frac{1}{|f|\rho} \left| \frac{\partial P}{\partial n} \right| = \frac{g}{|f|} \left| \frac{\partial Z}{\partial n} \right| & u_g &= -\frac{1}{f\rho} \frac{\partial P}{\partial y} = -\frac{g}{f} \frac{\partial Z}{\partial y} & v_g &= \frac{1}{f\rho} \frac{\partial P}{\partial x} = \frac{g}{f} \frac{\partial Z}{\partial x} \\
|\vec{V}_t| &= \frac{R}{|f|} \left| \frac{dT}{dn} \right| \left(\ln \left(\frac{P_1}{P_2} \right) \right) & v_t &= \frac{R}{f} \frac{\partial T}{\partial x} \left(\ln \left(\frac{P_1}{P_2} \right) \right) & u_t &= -\frac{R}{f} \frac{\partial T}{\partial y} \left(\ln \left(\frac{P_1}{P_2} \right) \right) \\
\frac{\zeta + f}{D} &= \text{constant} & f &= 2\Omega \sin \phi & \Omega &= \frac{2\pi}{(24)(60)^2} \text{s}^{-1} & \text{Div} &= \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) & \zeta &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \\
\Delta y &= \frac{\pi R_E}{180^\circ} \Delta \phi & \Delta x &= \frac{\pi R_E \cos \phi}{180^\circ} \Delta \theta & R_E &= 6.4 \times 10^6 \text{m} \\
c_R &= |\vec{V}_g| - \frac{\lambda^2 \Omega \cos(\phi)}{2\pi^2 R_E} & \lambda_c &= \pi \sqrt{\frac{2|\vec{V}_g| R_E}{\Omega \cos \phi}} & k\lambda &= 2\pi = 360^\circ
\end{aligned}$$